

KUru

Intelligent PLO-to-Career Navigator

An AI-powered academic pathway advisor
for students exploring programs at
Kasetsart University

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Powered by



Curriculum Data



RAG + LLM
AI Engine



Neo4j
Knowledge Graph

Students Exploring Programs at Kasetsart University

Connecting student interests to KU program PLOs,
TCAS pathways, and real graduate outcomes — all in one place.

Scope: pre-admission exploration · Thai & English

What KUru does and doesn't

IN SCOPE

- Interest discovery — find the right KU program
- Program recommendation — all KU programs
- PLO spider chart — visualize graduate skills
- TCAS guide — all rounds, all KU faculties (advisor data)
- Curriculum chatbot — Q&A over UAG.2
- Graduation data — first job & salary (advisor-provided)
- Saved profile — optional login

OUT OF SCOPE

- Skill tracker for enrolled students
- Elective recommender for enrolled students
- Career readiness report (Phase 2)
- Mobile app — web-only for MVP
- UAG.3 CLO-level data
- Real-time TCAS quota tracking
- Portfolio scorer

Three problems no existing system solves for KU

01

PLO documents are inaccessible

Curriculum is written for accreditation reviewers dense Thai PDFs that no student reads. Yet they contain everything students need to choose the right program.

100+ KU programs

buried in PDFs

02

TCAS guidance is fragmented

4 rounds, different score requirements per faculty, different portfolios per round. One misstep costs a student an entire year. Advisor-provided data eliminates risk.

~50,000+

TCAS applicants / year

03

No graduate outcome visibility

Students cannot find out what KU graduates actually do, first job titles or salary ranges, making it impossible to justify a program choice to parents.

4 years

cost of wrong choice

What KUru sets out to achieve



Surface hidden curriculum

Make PLO content from UPO.2 readable by students in Thai and English.



Personalized program match

Semantic similarity + knowledge graph maps interests to the right KU program.



Demystify TCAS admission

KU-specific round requirements, score thresholds, deadlines from advisor data.



Show graduate outcomes

First job titles and salary ranges from KU graduates with real evidence for decisions.



Grounded curriculum chatbot

RAG chatbot answers from real UPO.2 content with citations, no hallucinations.



KU-specific, not generic

Built exclusively for Kasetsart University, not a general career chatbot.

Why AI

Rule-Based / Manual

AI Approach — RAG + Gemini + Neo4j

Curriculum PDFs: inconsistent formatting across 100+ faculties

LLM + RAG understands context regardless of formatting

Student queries are natural language — semantic intent

Semantic embeddings capture intent, not just keywords

Interest → PLO → Skill → Career is multi-hop reasoning

Neo4j graph traversal handles typed multi-hop queries

Thai + English cross-lingual retrieval required

gemini-embedding-001 multilingual. one model, both languages

AI is necessary because PLO content requires understanding of Thai academic text — a rule-based system cannot do this.

Existing Tools and Their Limitations

ChulaGENIE

HIGH GAP

AI Career Advisor — Chulalongkorn University

ChulaGENIE

เส้นทางสู่การเป็น Data Engineer ที่จุฬาฯ

สนใจด้าน Data Engineering เหนือ! นี่เป็นอาชีพที่มีความต้องการสูงและมีเงินเดือนในยุคนี้อยู่สูงมาก มาดูเส้นทางการเรียนและพัฒนาทักษะกันค่ะ

Data Engineer คืออะไร?

Data Engineer เป็นผู้เชี่ยวชาญด้าน:

- 🛠️ สร้าง Infrastructure สำหรับจัดเก็บและประมวลผลข้อมูล
- 🔗 พัฒนา Pipeline เพื่อขนส่งและแปลงข้อมูล
- 🗄️ ดูแลระบบฐานข้อมูล (Database Management)
- ⚙️ ทำให้ข้อมูลพร้อมใช้ สำหรับ Data Scientists และ Analysts

หลักสูตรที่เหมาะสมที่จุฬาฯ

1 หลักสูตรที่เหมาะสมที่สุด

Limitations

- Only specific for Chula’s students
- No PLO-to-skill mapping; no page citations; answers unverifiable
- Chula-specific only — no KU programs, no TCAS for KU faculties

How KUru is better

- Every answer RAG-grounded in official KU curriculum with exact citations
- Structured PLO → Skill → Career knowledge graph — not generic Q&A
- KU-specific TCAS guide and graduate outcome data

Existing Tools and Their Limitations

Knowva

HIGH GAP

AI Curriculum Chatbot — KMITL

Limitations

- Admission chatbot only — answers not curriculum-grounded
- No skill mapping, gap analysis, or career matching output
- Not specific for KU admission and curriculum

How KUrU is better

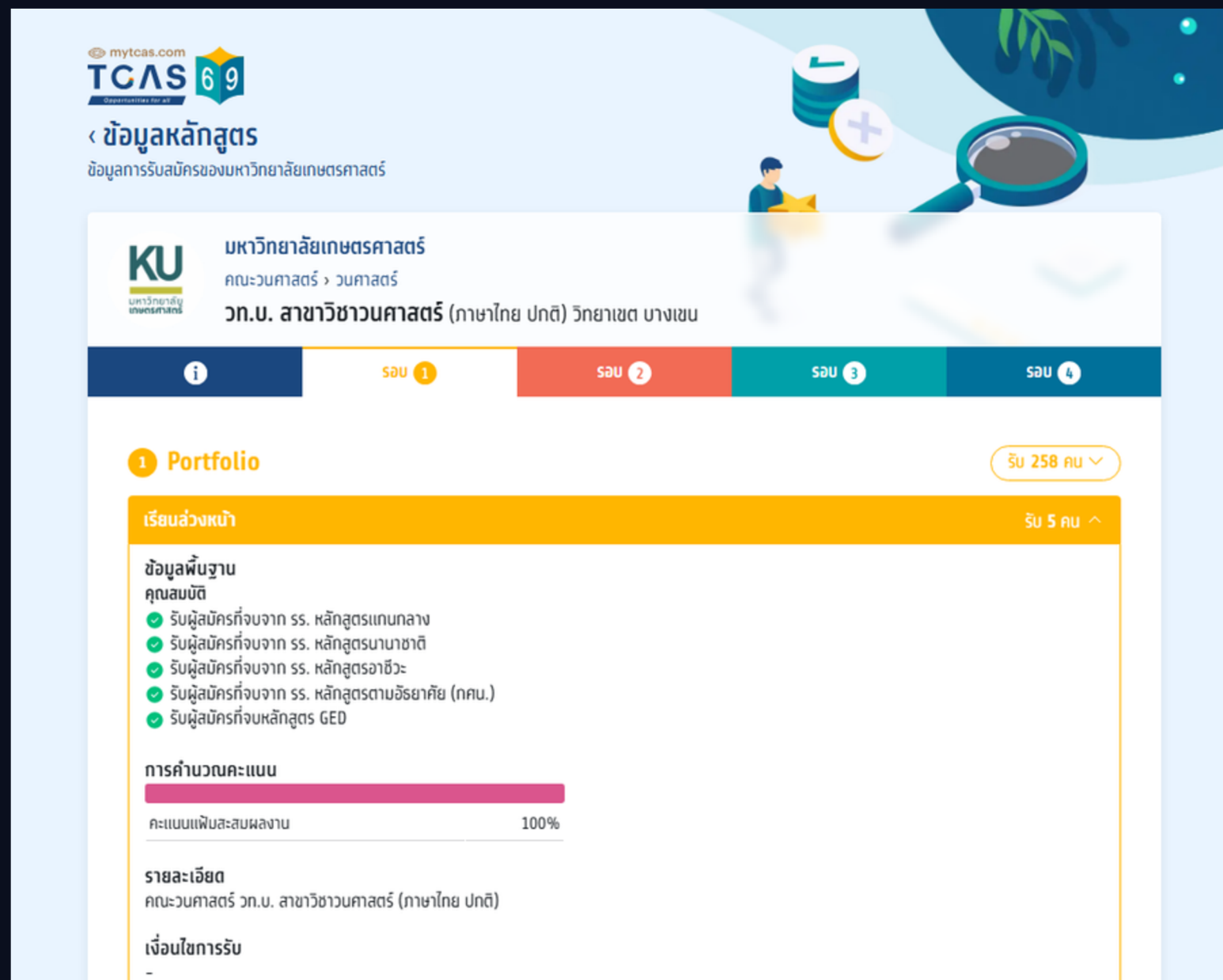
- Full RAG pipeline over มคอ.2 — every answer traceable to source
- PLO spider chart shows students exactly which skills each program builds
- Covers all KU faculties, not just one institution

Existing Tools and Their Limitations

TCAS Official Website

HIGH GAP

Static content, no personalization



Limitations

- Information scattered across dozens of faculty pages — no central navigation
- PLOs buried in dense Thai PDF documents; no plain-language explanation
- No career mapping, no graduate outcome data

How KUru is better

- All KU program information surfaced in one conversational interface
- PLOs translated to plain-language skill descriptions students can actually read
- TCAS guide and graduation data shown alongside program recommendations

Existing Tools and Their Limitations

General ChatGPT / Gemini

MED GAP

General-purpose AI assistants (not KU-specific)

Limitations

- No access to actual UAF.2 content — hallucinates PLOs and TCAS details
- Cannot cite specific curriculum sections; unverifiable answers
- No persistent memory; not tailored to KU programs or Thai TCAS system

How KUru is better

- Grounded strictly in official KU documents — zero hallucination on PLO content
- Every answer cites the exact UAF.2 section it came from
- Persistent saved profile and session memory for returning students

How KUru differs from existing solutions

Feature	KU Website	ChulaGENIE	Knowva	MyTcas	KUru
Grounded in official KU curriculum	✓	✗	✗	✗	✓
PLO explanation in plain language	✗	✗	✗	✗	✓
Skill radar chart (PLO-based)	✗	✗	✗	✗	✓
TCAS guide	✓	✓	✓	✓	✓
Graduate outcome data (salary, job)	✗	✓	✗	✓	✓
Thai & English support	✗	✓	✗	✓	✓
Source citations for every answer	✗	✗	✗	✗	✓
KU-specific scope	✓	✗	✗	✗	✓

Data Sources and Availability

KU Curriculum Documents

CONFIRMED

Provider: Faculty advisor · Format: Thai-language PDFs · Coverage: All KU programs

Authoritative source for PLOs, course content, and learning outcomes. Provided directly — no scraping required.

PyMuPDF → chunk by structure → embed (text-embedding-004) → pgvector + Neo4j

TCAS Admission Data

CONFIRMED

Provider: Faculty advisor · Format: Structured dataset · Coverage: All KU faculties, all rounds

Score thresholds, portfolio requirements, deadlines per faculty. Advisor-provided — no annual refresh risk.

Structured ingest → PostgreSQL → TCAS guide feature

Graduation Outcome Data

CONFIRMED

Provider: Faculty advisor · Format: Structured dataset · Coverage: KU graduates per program

First job titles and salary ranges per program. Shown in Graduation Data Showcase for real evidence.

Structured ingest → PostgreSQL → Graduation Showcase feature

All three datasets confirmed by the faculty advisor — zero data availability risk.

How KUru's AI works

RAG Pipeline – Curriculum Chatbot

Student query

Thai or English natural language



Embed query

gemini-embedding-001



pgvector search

Cosine similarity → top-k chunks



Prompt assembly

Context + history + query



Gemini 2.5 Flash

Generate grounded answer



Answer + citations

Links to UAD.2 sections

Eval: RAGAS faithfulness > 0.80 · Answer relevancy > 0.75

Recommendation Pipeline

Interest profile

8-topic grid + follow-up answer



Skill cluster mapping

Weighted skill vector



pgvector similarity

Candidate program retrieval



Neo4j graph traversal

Faculty → PLO → Skill → Career



Fit score ranking

Ranked KU programs



LLM explanation

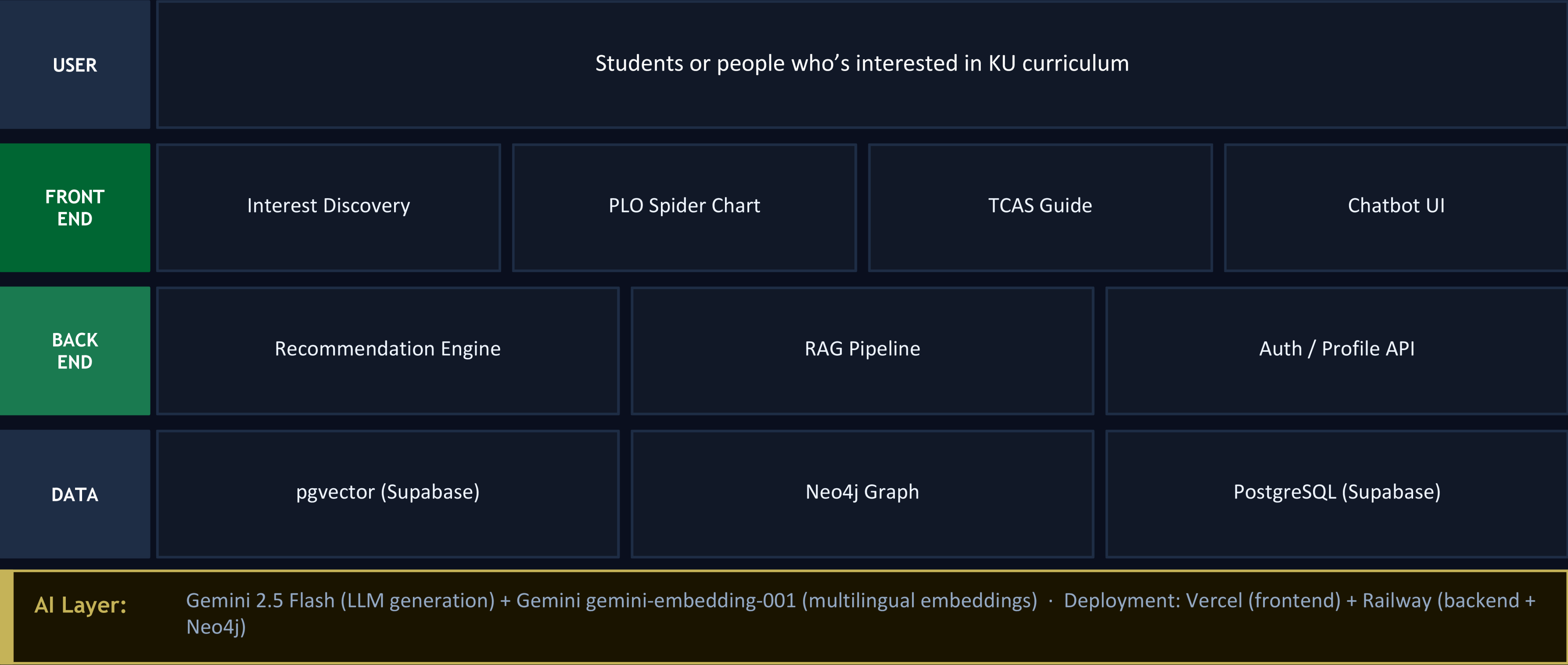
Why this program fits you

Eval: MRR / NDCG@5 > 0.60 · 30-50 student profiles

Model Choices and Evaluation Baseline

Gemini gemini-embedding-001 Embeddings	Multilingual Thai + English. Enables cross-lingual retrieval with Thai query finds English document chunks. No translation step needed.
Gemini 2.5 Flash LLM Generation	Strong Thai language capability, 1M token context window, native PDF ingestion, cost-efficient. Powers chatbot and recommendation explanation.
Neo4j + Supabase pgvector Data Layer	pgvector for semantic similarity (fuzzy matching), Neo4j for typed graph traversal (Faculty → PLO → Skill → Career). Each solves a different query type.
Baseline:	BM25 keyword retrieval + zero-shot Gemini (no RAG). Expected faithfulness ~0.40–0.50. KUru target: > 0.80 (RAGAS). Improvement will be measured and reported.

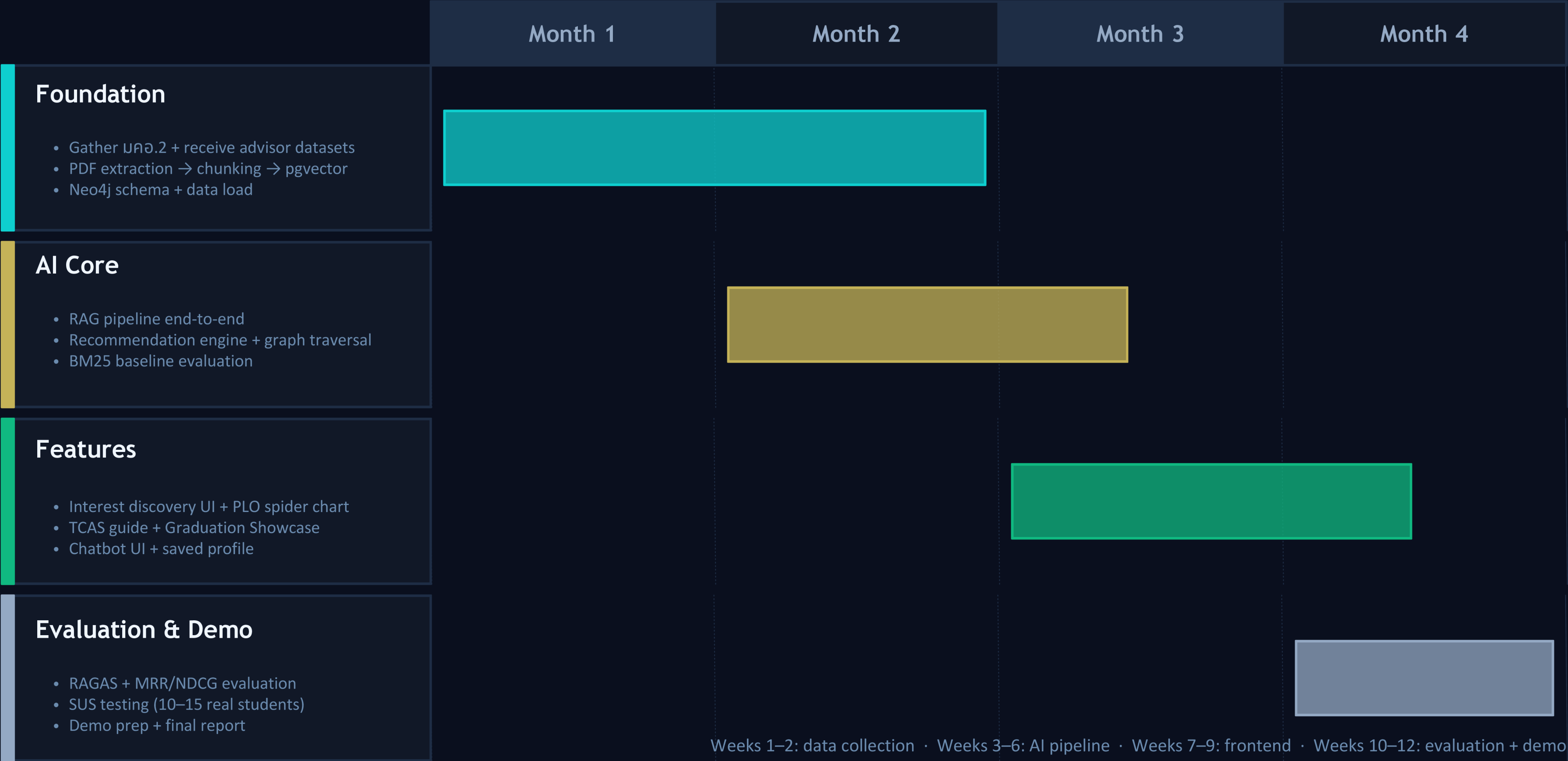
System Architecture



Success Metrics

Dimension	Metric	Target	How Measured
RAG Faithfulness	RAGAS Faithfulness	> 0.80	Automated RAGAS on Q&A test set from UAD.2. Measures: does the answer only use retrieved content?
RAG Relevance	RAGAS Answer Relevancy	> 0.75	Measures whether the generated answer addresses the student's actual question.
Recommendation Quality	MRR / NDCG@5	> 0.60	30–50 student profiles with ground-truth faculty matches, validated by faculty advisor.
Usability	SUS Score	> 70	User testing with 10–15 real pre-admission students. Structured task script. Acceptable = 70+.
Task Completion	Task Completion Rate	> 80%	End-to-end: Interest discovery → recommendation → TCAS guide → chatbot interaction.

4-month Development Plan



KUru

We turn the documents students never read

into a personal advisor that shows them where they are,
where they can go, and exactly what they need to get there.

Llama 2 + RAG

Gemini AI

Neo4j Graph

Next.js + FastAPI

Thai & English

KU-specific data

Thank you — Ready for questions

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